

Naming Files

File names are case sensitive

Restrict symbols to **dot(.)** **dash(-)** and **underscore (_)**

Filenames with spaces

escape the space `some\ file`

quotes `"some file"`

Dot Files

- Filenames can begin with a dot
- These files are hidden from view by most utilities
- They are mostly used for storing config files in home directory

single dot (.) filename - current directory

double dot (..) filename - parent directory

Wildcards

A symbol that stands for other characters

Question mark **?**

single character

`b??k` → book etc

Asterisk *****

any characters, including no character

`b*k` → bk, book

Brackets **[]**

any character is the set

`b[ae]ok` → baok

`b[ae]o[kds]` → baok

Range of values **[-]**

`b[a-z]ck` → back

Example

wildcards can be used in commands

```
james@laptop:~$ ls
Desktop  Downloads  IdeaProjects  Pictures  Templates
Documents  gitconfig  Music        Public   Videos
james@laptop:~$ ls g*g
gitconfig
```

Combining options

You can combine options e.g. `ls -lF`

ls

list

```
ls [options] [files]
```

a	show all files
l	long listing - includes permissions, ownership, file size

```
james@laptop:~/Pictures$ ls -al
total 20
drwxr-xr-x  2 james james 4096 Jun 29 19:37 .
drwxr-x--- 33 james james 4096 Jul 21 19:01 ..
-rw-rw-r--  1 james james 8945 Jun 29 19:37 trademe.png
```

cp

copy

```
cp [options] source destination
```

source can be one or more files

R	recursive - copy directory and its contents
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example

```
cp -R TempDir NewDir
```

mv

move

```
mv [options] source destination
```

- move files and directories
- can optionally rename them

Renaming a file/directory

example

```
mv document.txt sale.txt
```

rm

remove

```
rm [options] files
```

r R	recursive
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can't recover a removed file

touch**touch [options] files**

- update access and modification time of a file
- by default sets modification and access time to current time

Linux maintains 3 timestamps for every file:

1. last file-modification time
2. last inode change (inode stores file metadata)
3. last access time

-t timestamp	set time as specified
-a --time=atime	change only access time
-m --time=mtime	change only modification time

set timestamp syntax **MMDDhhmm[[CC]YY][.ss]**

hh - hour (24 hour clock)

[[CC]YY] - year e.g. 2012 or 12

ss - second

Archiving files

Archiving

- combining multiple files/directories into a single file, known as an **archive file**
- zip is also supported on Linux

Archiving Tools

tar - standard

cpio - not used much

dd - archive a file, device, filesystem (not used much)

tar

tape archive

`tar [options] destination/tar-filename files-to-archive`**tarball** is a archive file created by tar, it's then usually compressed

use one command with any option/s

commands

<code>-c --create</code>	create archive
<code>-t --list</code>	list archive contents
<code>-x --extract --get</code>	extract files from archive

options

<code>-f --file file</code>	specify the name of the archive you're creating - <i>must be followed immediately by the archive name</i>
<code>J --xz</code>	compress archive with xz

compression tools

gzip	low compression (fast)	.gz
bzip2	medium compression	.bz2
xz	high compression (slow) - best	.xz

example

compress files

```
james@laptop:~/Downloads$ tar -cJf my_documents.tar.xz Neurologist.pdf ENT.pdf
james@laptop:~/Downloads$ ls *.tar.xz
my_documents.tar.xz
```

my_documents.tar.xz**.tar** - file is a tarball**.xz** - file compressed using xz

compress a directory

`tar cJf documents.tar.gz ~/Downloads/`**dd**

data duplicator

copies files at low level

`dd if=<input-file> of=<output-file> [options]`**if** - can be a file, device, filesystem**of** - can be file, device or /dev/null (discards output)

to restore swap the if and of options

Directory Commands

mkdir

make directory

`mkdir [options] directory-names`

When deleting a directory tree filled with files, use **rm -r** instead of **rmdir**. This is because **rmdir** can only delete empty directories

rmdir

remove directory

`rmdir [options] directory-names`